

Module Reference

Pandoras Box -- User Documentation

Version 1.0 | pandoras-box.ai

Module Reference

Version: 1.0 **Audience:** Anyone selecting or managing modules

Disclaimer

This software is provided under the MIT License on an "as-is" basis, without warranty. Module costs are estimates only. You are responsible for verifying pricing with each third-party service provider. Nothing here constitutes financial advice.

How Modules Work

The core system includes everything needed to run company agents and a Personal Assistant. Modules are optional extensions that add specific capabilities.

To install a module:

```
sudo bash /opt/pandoras-box/scripts/add-module.sh
```

To remove a module:

```
sudo bash /opt/pandoras-box/scripts/remove-module.sh [module-name]
```

Removing a module stops the related services and removes their configuration. It does not delete any underlying data (emails, calendar events, documents).

Core Modules

core

Status: Required

The foundation. Installs the four-tier agent architecture, the SQLite job queue, the Security Overseer, and the base conductor and task agents for each company.

What breaks if not installed: Everything. All other modules depend on core.

Prerequisites:

- macOS 14 or later
- Node.js 20 or later
- Homebrew
- Anthropic API key

Monthly cost impact: Anthropic API usage (£10-50 depending on volume)

muse-browser

Status: Recommended

Your Personal Assistant with a full browser interface. Provides a unified view of all companies, morning briefings, research, conversation, and task management.

What breaks if not installed: You have company agents but no personal AI. Morning briefings, cross-company views, and conversational assistance are not available.

Prerequisites: core, Anthropic API key

Monthly cost impact: Additional Anthropic API usage (£5-20 depending on use)

Email Modules

mail-ms365

Status: Optional

Microsoft 365 email integration via Microsoft Graph API.

Enables Mail Agents to read, send, and search email using Outlook or Exchange.

What breaks if not installed: Company agents cannot handle email for Microsoft 365 companies. The Conductor still works -- it just cannot route mail jobs.

Prerequisites:

- core
- Microsoft 365 account
- Azure app registration (client ID, tenant ID, client secret)
- Microsoft Graph permissions: Mail.ReadWrite, Mail.Send

Monthly cost impact: None (Microsoft 365 subscription cost is separate)

mail-google

Status: Optional

Gmail integration via Google OAuth.

Enables Mail Agents to read, send, and search email using Gmail accounts.

What breaks if not installed: Company agents cannot handle email for Google Workspace companies.

Prerequisites:

- core
- Google account
- Google Cloud project with Gmail API enabled
- OAuth 2.0 client credentials

Monthly cost impact: None (Google Workspace cost is separate)

calendar

Status: Optional

Calendar integration. Reads existing events, creates new events, and manages invites. Auto-detects whether the company uses Microsoft 365 or Google Calendar.

What breaks if not installed: Calendar queries and meeting briefs are unavailable. Email and files still work.

Prerequisites: core, mail-ms365 or mail-google (at least one)

Monthly cost impact: Minor additional Anthropic API usage for calendar queries

files

Status: Optional

SharePoint (Microsoft 365) or Google Drive document access.

What breaks if not installed: Document queries and file management are unavailable. Email and calendar still work.

Prerequisites: core, mail-ms365 or mail-google (at least one)

Monthly cost impact: Minor additional Anthropic API usage for document queries

Interface Modules

admin-lite

Status: Optional

Mobile-friendly Admin Panel, accessible via Tailscale from any device.

Provides PIN-protected access to service status, logs, job queue, and command sending from a phone or tablet.

What breaks if not installed: Remote admin from mobile is unavailable. The Dashboard (local network) still works.

Prerequisites: core, Tailscale installed

Monthly cost impact: None

dashboard

Status: Optional

Local service status Dashboard. A web page on your local network showing all running services, recent job activity, and system health.

What breaks if not installed: No visual service overview. You can still check status via the terminal.

Prerequisites: core

Monthly cost impact: None

terminal

Status: Optional

Browser-based Terminal with passphrase authentication. Accessible on your local network.

What breaks if not installed: Browser terminal unavailable. Standard macOS Terminal app still works.

Prerequisites: core

Monthly cost impact: None

docs-server

Status: Optional

Local Documentation Server. Renders the Pandoras Box manuals as a navigable website on your local network.

What breaks if not installed: Manuals only accessible as raw Markdown files.

Prerequisites: core

Monthly cost impact: None

AI and Infrastructure Modules

ollama

Status: Optional

Local LLM for message classification. Runs on your Mac and handles high-volume tasks without calling the Anthropic API, significantly reducing costs.

The default model is `gemma3:12b`. Conductors detect Ollama automatically and fall back to the Anthropic API if it is unavailable.

What breaks if not installed: All AI processing uses the Anthropic API. Higher costs for high-volume usage. No functional difference in output quality.

Prerequisites:

- core
- 16 GB RAM minimum
- Ollama installed (`brew install ollama`)
- `gemma3:12b` model pulled (`ollama pull gemma3:12b`)

Monthly cost impact: Reduces Anthropic API costs by 30-70% for active use

athena

Status: Optional

Knowledge RAG (Retrieval-Augmented Generation). A vector store for document retrieval that enables agents to search a curated knowledge base for relevant context before generating responses.

Reduces AI hallucination for domain-specific or internal knowledge queries.

What breaks if not installed: Agents answer from training knowledge only. For general questions this is fine. For questions about your specific internal documents or policies, accuracy may be lower without Athena.

Prerequisites:

- core
- Qdrant or compatible vector store running and accessible
- ollama recommended (reduces embedding API costs)

Monthly cost impact: Minor additional API usage for embedding generation; Qdrant can run locally at no cost

Relay Modules

relay-discord

Status: Optional

Discord relay. Receive messages from a Discord server and reply through the same channel.

Prerequisites:

- core
- Discord bot token
- Discord server with the bot invited and appropriate permissions

Monthly cost impact: None

relay-slack

Status: Optional

Slack relay. Receive messages from Slack and reply via the same workspace.

Prerequisites:

- core
- Slack app with `chat:write` and `channels:history` scopes
- Slack OAuth token

Monthly cost impact: None

relay-whatsapp

Status: Optional

WhatsApp relay via an unofficial bridge.

Important: Verify compliance with WhatsApp platform terms of service before deploying in a production environment. Unofficial bridges may violate terms of service. Use at your own risk.

Prerequisites:

- core
- WhatsApp account
- Compatible bridge software installed and authenticated

Monthly cost impact: None

Specialist Modules

watch-companion

Status: Optional

Pixel Watch companion for notifications and quick replies.

Delivers morning briefing summaries and system alerts as watch notifications. Supports quick-reply actions for common commands.

Prerequisites:

- core
- muse-browser
- Android phone with companion app installed
- Pixel Watch paired to the phone

Monthly cost impact: None

clio

Status: Optional

Trading and investment signals module. Monitors a watchlist, tracks technical and event-driven signals, and surfaces opportunities in your morning briefing.

Important: Clio is not a financial advice service. All signals require human review before action. Trading involves significant risk of loss. You are responsible for all trading decisions and their outcomes.

Prerequisites:

- core
- Brokerage or market data API credentials
- Funded account with a supported data provider

Monthly cost impact: £10-20 additional Anthropic API usage due to signal processing frequency

calliope

Status: Optional

Social media and content publishing pipeline. Produces drafts, manages a content queue, and publishes on instruction after your approval.

What it requires for publishing:

- API credentials for each platform you want to publish to

Monthly cost impact: £5-15 additional Anthropic API usage depending on publishing volume

urania

Status: Optional

Agent self-improvement pipeline. Analyses agent performance logs on a weekly schedule, generates prompt adjustment proposals (GEPA), and applies approved proposals automatically.

Also includes a skill library with a 72-hour review interval.

Over time, Urania helps your agents improve their accuracy and relevance based on actual usage patterns -- without requiring manual prompt engineering.

Prerequisites: core, muse-browser

Monthly cost impact: Minor (weekly review batch, not continuous)

video-publisher

Status: Optional

Automated video production and YouTube publishing.

Combines script generation, voice synthesis (ElevenLabs), and video assembly (ffmpeg) to produce and publish videos automatically. Integrates with the Calliope content queue for scheduling.

Prerequisites:

- core
- ElevenLabs API key
- ffmpeg installed (`brew install ffmpeg`)
- YouTube Data API v3 credentials with upload permissions
- calliope recommended for content queue integration

Monthly cost impact: ElevenLabs voice synthesis cost depends on script length (typically £1-5 per video); Anthropic API for script generation (£2-5 per video)

UI Modules in Detail

The following sections describe the user interface modules in more detail.

Admin Shell

The Admin Shell is a Chrome desktop application. It opens on your Mac like any other app.

How to access: Click the Admin Shell icon in your Dock or Applications folder.

What it looks like: A dark-themed browser window with tabs for:

- Status (live service health, green/amber/red indicators)
- Logs (searchable log viewer, all services)
- Queue (job queue with status filters)
- Projects (autonomous build project list)
- Deploy (deploy controls, DRY_RUN toggle)

What you can do:

- See at a glance which services are running
 - Search logs for errors across all services
 - Monitor job queue activity
 - Review and approve build projects
 - Run a deploy
-

Dashboard

The Dashboard is a web panel accessible on your local network.

How to access: Open a browser on any device on your home or office network and go to

`http://[your-mac-hostname].local:8181`

Example caption: Example: Greek mythology theme shown.

What it looks like: A dark dashboard with service tiles, a recent jobs feed, and a system health summary.

What you can do:

- Quick service status overview
- See recent job activity across all companies
- No controls -- read-only view

Terminal

The browser-based Terminal is a command-line interface in a browser tab.

How to access: `http://[your-mac-hostname].local:8282`

What it looks like: A classic terminal emulator in a browser window. Black background, monospace font.

Authentication: Passphrase required on each session.

What you can do: Anything you can do in the macOS Terminal -- run commands, check logs, restart services.

Admin Panel (Lite)

The mobile-friendly Admin Panel is gated behind Tailscale.

How to access: `https://[your-tailscale-address]:8787` from any Tailscale device.

What it looks like: A minimal, touch-friendly interface optimised for phone screens.

Authentication: PIN code required (5 failed attempts triggers a 15-minute lockout).

What you can do:

- Check service status
 - Restart a service
 - View recent jobs
 - Send a command to an agent
 - View the last 50 log lines per service
-

Personal Assistant Browser Interface

The full Personal Assistant browser interface.

How to access: `https://[your-tailscale-address]:8800`

What it looks like: A full-featured chat interface with a sidebar showing recent conversation history, quick-action buttons, and a settings panel.

Example caption: Example: Greek mythology theme shown.

What you can do:

- Full conversation with your Personal Assistant
 - Upload files for discussion (drag and drop)
 - Access conversation history and search past interactions
 - Configure briefing settings
 - View and manage memory
 - Access links to your company agents
-

Documentation Server

A local web server that hosts these manuals.

How to access: `http://[your-mac-hostname].local:8485`

What it looks like: A clean documentation site with navigation for all manuals.

What you can do: Read any manual in a formatted, searchable browser interface.